



Research paper

A novel crude oil price forecasting model using decomposition and deep learning networks

Yao Dong^a, He Jiang^{b,*}, Yunting Guo^a, Jianzhou Wang^c^a School of Statistics and Data Science, Jiangxi University of Finance and Economics, Nanchang Jiangxi 330013, China^b School of Economics and Finance, Xi'an Jiaotong University, Xi'an 710061, China^c The Institute of Systems Engineering, Macau University of Science and Technology, Taipa Street, Macau

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ABSTRACT

Crude oil prices are susceptible to a variety of elements, including military, political, financial, and diplomatic forces, resulting in price fluctuations that are random, abrupt, and chaotic. Based on the chaotic nature of crude oil price series, this paper proposes a model for predicting crude oil prices that incorporates variational modal decomposition algorithm (VMD), phase space reconstruction technique (PSR), convolutional neural network (CNN), and bidirectional long and short-term memory network (BiLSTM). Specifically, the noises in the original data are removed using VMD. Next, the crude oil price is reconstructed using PSR. Then the reconstructed and denoised phase space is fed into the CNN-BiLSTM model for multi-step predictions. The empirical results show that the proposed model achieves the lowest MAPEs as 0.662%, 2.283%, 3.829%, and 4.44% and the lowest MSEs as 0.335, 3.942, 10.047 and 13.681. The Diebold Mariano (DM) test also demonstrates the proposed model's superiority in the domain of crude oil price forecasting.

1. Introduction

Crude oil is an essential nonrenewable fossil fuel for a nation's political stability and economic development (Baz et al., 2021). Crude oil prices are extremely volatile and are susceptible to various uncertainty factors including military, political, economic, and diplomatic forces (Fan and Xu, 2011; Van Eyden et al., 2019). For example, international crude oil prices have plummeted since the outbreak of COVID-19 pandemic in 2020. Due to the Russia–Ukraine conflict, the Brent crude oil spot price averaged \$97 a barrel in February 2022, up \$11 a barrel from January, while the West Texas Intermediate (WTI) crude oil spot price averaged \$91.64 a barrel, up \$8 a barrel from January. For the first time since September 2014, Brent crude oil prices closed above \$100 per barrel when they reached nearly \$124 during the first week of March. Effective forecasting of crude oil prices can help relevant government staff and financial investors take timely and effective action to protect the country's economy from oil price shocks ahead of dramatic fluctuations in crude oil prices. So how should one construct an accurate crude oil price forecast? Despite the reality that global oil prices are very volatile and difficult to estimate, academics have been able to create numerous research findings in this area. Some scholars employed regression models based on statistical approaches to provide projections that were close to the real price change curve, including the autoregressive integrated moving average

model (ARIMA) (Zhou and Dong, 2012), generalized autoregressive conditional heteroskedasticity model (GARCH) (Hou and Suardi, 2012), quantile regression model (QR) (Sun et al., 2021) and hybrid statistical regression model (i.e., ARMA-FIGARCH-M) (Lin et al., 2020). However, crude oil prices are often non-stationary time series, making it challenging to obtain acceptable prediction accuracy using time series based models. Furthermore, this method depends on linear and normal distribution assumptions, which do not adequately reflect the crude oil price series' specific characteristics.

To tackle this challenge, machine learning methods are taken into consideration. Ding (Ding, 2018) proposed an alternative model for predicting the crude oil prices based on the ensemble empirical mode decomposition (EEMD) and artificial neural network (ANN). Abdollahi (Abdollahi, 2020) constituted a hybrid technique that included EEMD, support vector machine (SVM), particle swarm optimization (PSO), and Markov-switching generalized autoregressive conditional heteroskedasticity for forecasting. Zhang and Zhang (2018) proposed a new hybrid forecasting approach based on Hidden Markov, exponential generalized autoregressive conditional heteroskedasticity and least squares support vector machine model (HMM-EGARCH-LSSVM), which was also more accurate and effective in predicting crude oil prices for a long time period.

* Corresponding author.

E-mail address: jiangsky2005@aliyun.com (H. Jiang).<https://doi.org/10.1016/j.engappai.2024.108111>

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Traditional regression methods are less effective in forecasting non-stationary and non-linear time-series data, while machine learning models are unable to match the prediction requirements due to their tendency to overfit. The prediction of crude oil prices now frequently uses deep learning techniques. Scholars tend to propose hybrid models with data-driven approaches for forecasting. Wang et al. (2018) validated empirical forecasts on daily, weekly, and monthly crude oil price data using a novel hybrid method that combined an integrated data fluctuation network (DFN) and several artificial intelligence algorithms. To forecast crude oil prices, Busari and Lim (2021) combined the adaptive boosting algorithm with the long short-term memory network (AdaBoost-LSTM) and the gate recurrent unit network (AdaBoost-GRU). The benchmark models were set to a single LSTM and a single GRU, the AdaBoost-GRU model produced the best forecasting results. Zhang et al. (2021) proposed a hybrid modeling approach for analyzing crude oil price volatility based on EEMD, variational mode decomposition (VMD) and the kernel extreme learning machine (KELM) model optimized by particle swarm optimization (PSO). Hu (2021) used a combination of complete ensemble empirical mode decomposition (CEEMDAN), LSTM network, and attention mechanism to predict crude oil prices, which also constructed a news sentiment index based on Chinese oil news content. The experimental results demonstrated the effectiveness of the model. The above hybrid deep learning model extracts the non-linear features in the data better, but the prediction accuracy needs to be improved because of the complexities of crude oil price data. Li et al. (2021) studied a novel scheme by combining VMD and random sparse Bayesian learning (RSBL) to forecast the crude oil price. Wang et al. (2022) proposed a novel carbon price forecasting model based on a multi-source information fusion (MSIF) and hybrid multi-scale decomposition (HMSD).

In summary, the existing literature fully demonstrates the superiority of hybrid deep learning models combined with data-driven methods in the field of crude oil price forecasting. The general idea of this method is to first decompose the data into several frequency patterns, input the results to different prediction models according to the data characteristics of different patterns, and finally aggregate them. But it also has some flaws: First, forecasting typically begins with a one-dimensional time series. Because the data patterns are established, the prediction patterns are fixed, and no additional research can be done. Second, while de-noising aggregation is frequently employed with data-driven approaches, not all data is suitable for such procedures, and basic denoising may produce the best results. Third, the predictive capability of a hybrid model is determined by the combination of models, and each component should play a certain role.

To fill these gaps, the noteworthy aspect of this paper is the application of chaos theory to crude oil price prediction, which breaks the limitation of prediction only in one-dimensional time domain. In terms of mathematics, a nonlinear system is chaotic when it is sensitive to the initial and topological mixing and has dense periodic orbits (Bao et al., 2020). Based on chaos theory, a one-dimensional time series is mapped to the multi-dimensional phase space, which can extract information about the regularity and dynamics of the data in the original system to improve the forecasting performance. Meanwhile, a chaotic system is unpredictable in the long term due to the initial sensitivity law. Hu (2021), Li et al. (2021) and Wang et al. (2022) predicted the crude oil price series, but in fact, it should be preceded by an analysis of whether the nonlinear system is chaotic. With the continuous development of chaos theory, it has attracted the attention of scholars in economy (Kazem et al., 2013), industry (Meng et al., 2017), power engineering (Chen et al., 2022), energy (Sun and Zhang, 2022) and many other fields. To the best of the authors' knowledge, there is little literature on crude oil prices from a chaos theory perspective. The studies of HE (2011), Lahmiri (2017), Jozmaleki et al. (2020), Niu et al. (2020) provided evidence of crude oil market chaos, but did not consider crude oil price forecasts. Yin and Wang (2019) considered the chaotic characteristics of crude oil price time series and used the back

propagation network (BP) to forecast crude oil prices, however, the forecasting accuracy is not satisfactory.

It is also worth noting that the data in this paper is decomposed and denoised using a novel data-driven approach to further improve prediction ability and an appropriate mixed forecasting model is selected to make the prediction. The main contributions of this paper are given as follows:

- (1) The time-frequency analysis method of decomposing and reorganizing data is a common data-driven method to improve the generalization ability of models. A new adaptive signal processing method variational modal decomposition (VMD) is applied in this paper when performing data processing, which can handle non-linear, non-stationary signals more efficiently and avoid endpoint effects and spurious component problems encountered during iterations compared to other time-frequency analysis methods. The high-frequency noises in crude oil prices are eliminated using such method, which improves prediction accuracy.
- (2) While general time series are studied primarily in the time domain, chaotic time series are forecast in the phase space. To compensate for the scarcity of traditional time-domain analysis methods, this can extract representative nonlinear feature information that is difficult to extract in one-dimensional space in high-dimensional phase space. In this paper, phase space reconstruction (PSR) is performed first in the data processing. In PSR, the choice of time delay (τ) and embedding dimension (m) is essential. The autocorrelation coefficient method is used to solve τ and m is determined by the false nearest neighbors method. Meanwhile, the maximum Lyapunov exponent is used to verify the chaotic nature of the sequence.
- (3) In the realm of time series forecasting, hybrid deep learning models routinely outperform single models, which frequently use neural networks. In chaotic time series prediction, different types of neural networks play diverse roles. The reconstructed phase space represents both the spatial and temporal features of the crude oil price chaotic system. Convolutional neural network (CNN) is a deep network model that integrates convolutional processing. The translation invariance of the CNN network ensures that the signal spectrum is retained and spatial information is extracted effectively during feature extraction. Bidirectional long short-term memory network (BiLSTM) is an improvement of LSTM network that works in both directions. It considers not only the time-series properties of the current time series but also the impact of future states on the present, as an excellent time-series prediction network. The combination of CNN and BiLSTM can improve the accuracy of the model prediction.

As previously stated, a VMD-PSR-CNN-BiLSTM model is proposed in this paper to forecast chaotic time-series crude oil prices. In terms of empirical evidence, this paper uses Brent crude oil spot price data for multi-step forecasting of one day, one week, half a month and 19 days. Finally, to assess the model's feasibility and superiority, the experimental results are compared to the unreconstructed, unnoised, and unprocessed model. The organization of this paper is given as follows: Section 2 presents the methodology of the proposed model. Section 3 shows the empirical analysis and a conclusion is drawn in Section 4.

2. Methodology

In this section, the hybrid model proposed in this paper will be introduced. The data processing method is introduced in Section 2.1. The data processing method includes two operations: noise reduction and phase space reconstruction. In particular, we first apply VMD to remove the noises with high frequency and the remaining part is used to construct the phase space using PSR. After that, a CNN-BiLSTM model is constructed using the phase space series as inputs. The CNN-BiLSTM model takes the advantages from both CNN and BiLSTM. This part will be introduced in Section 2.2. Finally, the flowchart of the proposed model VMD-PSR-CNNBiLSTM is described in Section 2.3.

2.1. Data processing approach

2.1.1. Variational modal decomposition

Variational modal decomposition algorithm (VMD) (Dragomiretskiy and Zosso, 2013) is a new time–frequency analysis method, which not only disintegrates a multi-component signal into multiple single-component AM and FM signals at one time, but also avoids the endpoint effect and false component problems encountered by in the iterative process. The solution process of the variational problem is the decomposition process of VMD. The intrinsic mode function (IMF) is defined as a bandwidth-limited AM-FM function in this approach. The VMD algorithm decomposes the original signal into a defined number of IMF components by constructing and solving the restricted variational problem. The VMD algorithm can be viewed as solving the following unrestricted problem.

$$L(m_k, c_k, \lambda) = \eta \sum_k \left\| \partial_t [(\delta(t) + \frac{j}{\pi t}) \cdot m_k(t)] \cdot e^{-j c_k t} \right\|_2^2 + \left\| \partial f(t) - \sum_k m_k \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_k m_k \right\rangle \quad (1)$$

where $\delta(\cdot)$ is the Dirac distribution. The symbols m and c , respectively, stand for the mode set and the corresponding center frequency. The k th mode decomposed is denoted by m_k . The original time series to be divided is represented by $f(t)$. η and λ denote quadratic penalty and Lagrangian multipliers, respectively. The specific algorithmic method is as follows:

- (1) Initialize $m_k^1, c_k^1, \lambda_k^1$ and $n = 0$. n is the number of iterations;
- (2) $n = n + 1$, enter the loop;
- (3) The inner loop is updated according to (2) and (3) until the number of decompositions reaches K ;

$$\widetilde{m}_k^{n+1}(c) = \frac{\widehat{f}(c) - \sum_{i \neq k} m_i^{n+1}(c) + \frac{\widehat{\lambda}^n(c)}{2}}{1 + 2\alpha(c - c_k)^2} \quad (2)$$

$$c_k^{n+1} = \frac{\int_0^\infty c |\widetilde{m}_k^{n+1}(c)|^2 dc}{\int_0^\infty |\widetilde{m}_k^{n+1}(c)|^2 dc} \quad (3)$$

- (4) Update λ according to (4);

$$\widehat{\lambda}^{n+1}(c) = \widehat{\lambda}^n(c) + \tau(\widehat{f}(c) - \sum_k |\widetilde{m}_k^{n+1}(c)|) \quad (4)$$

- (5) Given precision ε , if the following condition

$$\frac{\left\| \widetilde{m}_k^{n+1} - \widetilde{m}_k^n \right\|_2^2}{\left\| \widetilde{m}_k^n \right\|_2^2} < \varepsilon \quad (5)$$

is satisfied, stop the loop, otherwise enter (2) continue the loop.

2.1.2. Phase space reconstruction

The crude oil market is chaotic, which means that it is a non-linear dynamic system with chaotic properties. The initial stage in understanding chaotic dynamical systems is phase space reconstruction (PSR). Takens (1981) in 1981 showed that an appropriate embedding dimension m can be determined. The chaotic attractor can be reconstructed in this embedding dimension space if the delay coordinate's dimension satisfies $m \geq 2d + 1$ and the dynamic system's dimension is denoted by d .

Let the univariate time series be $\{x(t)|t = 1, 2, \dots, n\}$, adopting the commonly used coordinate delay reconstruction method (Packard et al.,

1980), the delay vector and trajectory matrix can be obtained after phase space reconstruction which is given as:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix} = \begin{bmatrix} x(1) & x(1+\tau) & \cdots & x(1+(m-1)\tau) \\ x(2) & x(2+\tau) & \cdots & x(2+(m-1)\tau) \\ \vdots & \vdots & \ddots & \vdots \\ x(n-1-(m-1)\tau) & x(n-(m-2)\tau) & \cdots & x(n-1) \end{bmatrix}, \quad (6)$$

where $N = n - (m-1)\tau$ is the number of phase points in the phase space, m is the embedding dimension, τ is the time delay and X_i is the phase point in phase space. According to Taken's theorem (Takens, 1981), if appropriate m and τ are chosen, the phase space can recover the attractor's dynamic features in the sense of topological equivalence. That is, the primary mover system can be recreated in a certain fashion under certain conditions. To confirm these two parameters, this research uses the autocorrelation coefficient method and the false nearest neighbor method, respectively.

The autocorrelation coefficient method is proposed in 1994 (Rosenstein et al., 1994). The primary idea behind this method for determining the delay time is to extract the linear correlation between the sequences. The autocorrelation function for a chaotic sequence $\{x(t)|t = 1, 2, \dots, n\}$ can be represented as follows:

$$A_{xx}(t) = \frac{1}{n} \sum_{t=1}^n x(t)x(t+\tau), \quad (7)$$

where the autocorrelation function value $A_{xx}(t)$ drops to $1 - e^{-1}$ of the initial function value, the corresponding τ is the delay time parameter.

When two non-adjacent points in high-dimensional phase space are projected onto a one-dimensional space axis, they are referred to as two adjacent points, or false neighbors. Recovering the chaotic motion's trajectory from a chaotic time series is the aim of reconstructing the phase space. The chaotic motion's trajectory will progressively open as m rises, and the false neighbors will gradually fade away, allowing the chaotic motion's trajectory to be retrieved. This idea serves as the foundation for the false nearest neighbors approach (Kennel et al., 1992).

For each vector in the m -dimensional phase space $X_{i(m)} = \{X_i, X_{i+\tau}, \dots, X_{i+(m-1)\tau}\}$, $i = 1, 2, \dots, N$, and N is the total number of vectors. Find its nearest neighbor vector $X_{j(m)}$, and calculate the Euclidean distance $R_m(i) = \|X_{i(m)} - X_{j(m)}\|_2$. The Euclidean distance between $X_{i(m)}$ and $X_{j(m)}$ in $m+1$ dimension space is given by:

$$R_{m+1}(i) = \|X_{i(m+1)} - X_{j(m+1)}\|_2. \quad (8)$$

If $R_{m+1}(i) \gg R_m(i)$, it is a false neighbor, and the ratio $R(i)$ is defined as:

$$R(i) = \sqrt{\frac{R_{m+1}^2(i) - R_m^2(i)}{R_m^2(i)}}. \quad (9)$$

Let R_0 be the threshold value which is usually greater than 10. If $R(i) \geq R_0$, then X_j is called the false neighbor of X_i . Calculate the percentage of false neighbors under m until the percentage of false neighbors is less than 5% as m increases. The required embedding dimension at this time is m .

2.1.3. The maximum Lyapunov exponent from small data sets method

The average exponential rate of convergence or divergence of the system over successive orbitals in phase space is represented by the Lyapunov exponent, which is a quantitative statistic that is used to quantify the dynamic features of the system. If the maximal Lyapunov exponent is greater than zero, the system has dynamic chaos. A positive Lyapunov exponent indicates that no matter how tiny the difference between the two initial paths is in the phase space of the system, the difference will grow exponentially with the passage of time, which is unpredictable. The maximum Lyapunov exponent from small data sets

method (Rosenstein et al., 1993) is used in this research to determine whether a time series is chaotic. The method using small data sets is a straightforward way of calculating the maximum Lyapunov exponent of chaotic time series, and its basic principle is as follows:

- (1) A fast Fourier transform is performed on the time series $\{x(t)|t = 1, 2, \dots, n\}$, and the mean period P is calculated;
- (2) After determining the time delay τ and the embedding dimension m , reconstruct the phase space $X_i|i = 1, 2, \dots, N$;
- (3) Find the nearest neighbor X_j of each point X_j in the phase space, and restrict the transient separation:

$$d_j(0) = \min_{X_j} \|X_j - X_j\|_2, \quad (10)$$

where $j - \hat{j} > P$ and $d_j(0)$ is the initial distance from the j th point to its nearest neighbor;

- (4) The geometric meaning of the maximum Lyapunov exponent λ_1 is a parameter that quantifies the evolution of the initial orbital characteristic with the exponential divergence. The sample period is denoted by Δt , and the distance between the j th and closest point pair on the basic orbit is denoted by $d_j(i)$. Then there are:

$$d_j(i) \approx C_j e^{\lambda_1(i+\Delta t)}, C_j = d_j(0) \quad (11)$$

where C_j is the initial separation, which is a constant. Take the logarithm of both sides of (11) yields that:

$$\ln d_j(i) = \ln C_j + \lambda_1(i \cdot \Delta t), \quad (12)$$

which depicts a collection of generally parallel lines and each with a slope that is roughly proportional to λ_1 .

- (5) Using a least-squares fit to the average line designated as (13), the maximal Lyapunov exponent can be computed quickly and precisely.

$$y(i) = \frac{1}{\Delta t} \langle \ln d_j(i) \rangle \quad (13)$$

where $\langle \ln d_j(i) \rangle$ denotes the average overall values of $\ln d_j(i)$. According to (13), the slope of the line is the maximum Lyapunov exponent λ_1 . The curves in this work are fitted using the random sample consensus algorithm (RANSAC) provided in Fischler and Bolles (1981).

2.2. Forecasting models

2.2.1. Convolutional neural network

A convolutional neural network (CNN) (Krizhevsky et al., 2012) has a powerful spatial feature extraction capability to identify local patterns of price sequences in phase space. The convolution layer works by convolving local regions of the input data with the convolution kernel, and traversing the entire input data by sliding the convolution kernel window to make the local sensory field. The phase space constituted of crude oil prices after PSR is the data to be handled in this research. The convolution kernel width will be set to the dimension of the phase space, and 1D-CNN will be used to extract the spatial features. One-dimensional means that the direction of the convolution is one-dimensional. The principle is shown in Fig. 1.

2.2.2. Bidirectional long and short-term memory network

Bidirectional long and short-term memory network (BiLSTM) (Huang et al., 2015) is a combination of forward LSTM and backward LSTM. To understand BiLSTM, we first introduce the LSTM. LSTM can learn the long-term dependencies of data. Its advantage is that the previous information is connected to the current task, and the current information is retained and passed on. The basic structure of LSTM neurons is shown in Fig. 2, where each unit has three gate structures, namely input gate i_t , forget gate f_t and output gate o_t . The update of the current input data to the state value of the memory cell is managed by

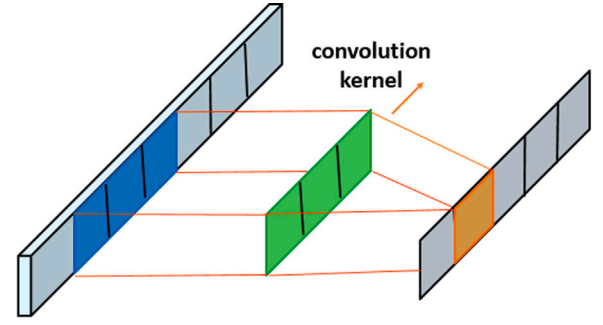


Fig. 1. One-dimensional Convolutional Neural Network.

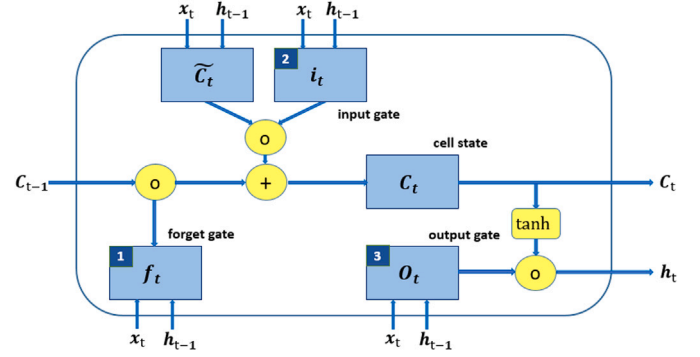


Fig. 2. Basic structure of LSTM neurons.

the input gate i_t . The update of the historical data to the state value of the memory cell is controlled by the forget gate f_t . The output state of the memory cell is managed by the output gate o_t . A chain of repeating units forms an LSTM layer.

At time t , x_t is the input value of the LSTM neuron, and h_{t-1} , h_t are the previous and current output values of the LSTM neuron, respectively. $\tilde{c}_t$ represents the storage unit of the current input data, and c_t indicates the state value of the current memory cell. A sigmoid function is used to express σ , while a hyperbolic tangent function is used to express $\tanh$. The calculation formula of LSTM neurons is as follows:

$$\begin{aligned} i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ \tilde{c}_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \\ c_t &= f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \\ h_t &= o_t \cdot \tanh(c_t) \end{aligned} \quad (14)$$

W_i, W_f, W_o, W_c represent the weight matrix of input gate, forget gate, output gate and memory cells. b_i, b_f, b_o, b_c represent the bias of input gate, forget gate, output gate and memory cells.

Time-related data can only be fitted in one direction using LSTM. BiLSTM incorporated forward LSTM layer and backward LSTM layer so that it might capture both the global and local patterns that LSTM could miss. BiLSTM can obtain important information from both past and future to enhance the forecasting accuracy. BiLSTM's structure is depicted in Fig. 3.

Suppose there are N inputs of time sequences X . At time t , the output Y_t of BiLSTM can be expressed as:

$$Y_t = h_t^f \oplus h_{N-t+1}^b \quad (15)$$

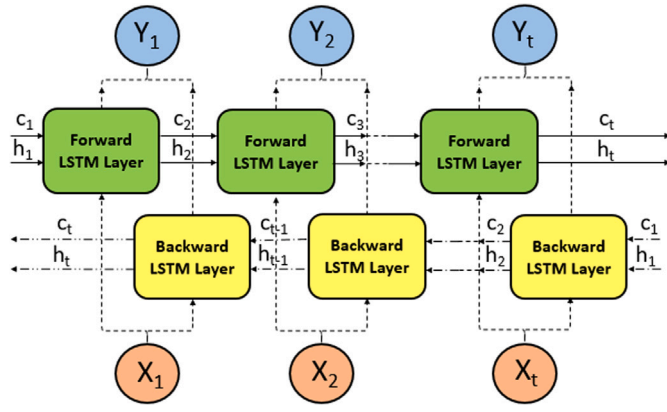


Fig. 3. The structure of BiLSTM.

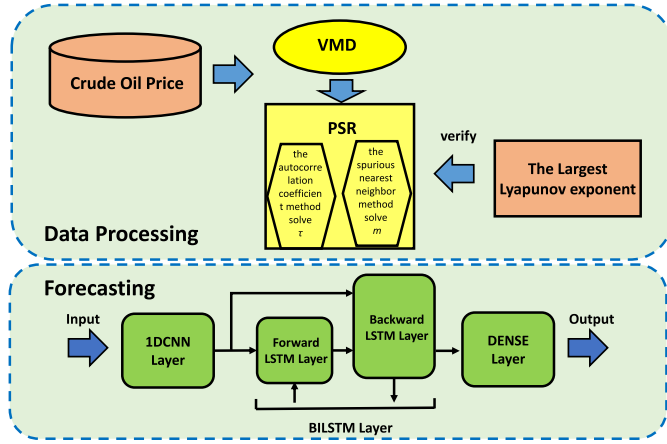


Fig. 4. The proposed VMD-PSR-CNN-BiLSTM model for crude oil price forecasting.

The outputs of the forward and backward LSTMs are h^f and h^b , respectively. Any integration operator, such as a basic adder or a neural network can be expressed using the notation $\oplus$.

2.3. The proposed hybrid model

The proposed hybrid model includes four steps and Fig. 4 illustrates the flow chart.

Step 1 Noise reduction. The crude oil price signal is decomposed into seven IMFs through the VMD algorithm. These IMFs are ordered from low frequency to high frequency and the highest-frequency IMFs which have high randomness and time variability will be removed as noise.

Step 2 Reconstruction. After removing the noise from the data, this study reconstructs the phase space of the crude oil price series, based on the chaotic character of the crude oil market. In this phase space, the regularity and order of the sequence are reflected, which facilitates the extraction of data features for prediction. The key step in reconstructing the phase space is the determination of time delay τ and embedding dimension m . To solve τ , the autocorrelation coefficient approach is employed, and to find m , the spurious nearest neighbor method is used.

Step 3 Chaos verification. When a phase space is reconstructed, if it does not have chaotic properties, then the existence of the phase space is unreasonable. The approach of calculating the maximum Lyapunov exponent from small data sets proves the

Table 1

Descriptive statistics of the data.

Dataset	Sample size	Mean	Std	Min	Max
Brent	5638	63.72	29.01	9.12	143.95
Training set	4510	64.00	31.66	16.51	143.95
Test set	1128	62.60	14.09	9.12	86.28

time series' chaotic nature. At the same time, the short-term predictability of chaotic systems limits the length of prediction time.

Step 4 Forecasting. Prediction of a space usually includes spatial positioning and temporal changes. The CNN-BiLSTM model is used to forecast the reconstructed phase space. The CNN model has its own advantages in extracting spatial features, and BiLSTM can simulate the changes well in the temporal dimension of the phase space. In order to verify the generality of the model, the prediction results of different time steps will be analyzed and discussed.

3. Empirical analysis

The empirical evidence in this paper includes both data processing and forecasting components. The data are first processed by the VMD method with the purpose of reducing noise and then are reconstructed in phase space. In the prediction section, the noise-reduced phase space is fed into the CNN-BiLSTM model for price prediction. In order to verify the validity of the model, the prediction result analysis and robustness analysis are also given in this paper.

3.1. Data description

Brent Crude, an international crude oil evaluation observation system, belongs to low sulfur crude oil. It is the most important and widely used oil price data in the world. Data on Brent crude spot prices is collected from the Global Economic Indicators website. (<https://tradingeconomics.com/>). In this paper, daily Brent crude oil spot prices from January 4, 2002, to December 31, 2021, are used as empirical data. The data is split to make the training set 80 percent of the data and the test set 20 percent of the data. Specifically, a total of 4510 data from January 4, 2000, to October 2, 2017, are used as the training set, and 1128 data from October 3, 2017, to December 31, 2021 are used as the test set. Table 1 shows descriptive statistics for the data, including sample size, mean, standard deviation, minimum, and maximum values. Because the data should be tensored into phase space, in fact, only a total of 365 data from October 2, 2020, to December 31, 2021, are predicted during the test. A total of 5638 data points were gathered, resulting in the crude oil price time series depicted in Fig. 5. It can be preliminarily judged that the crude oil price series is a non-stationary sequence, and the data needs to be preprocessed before forecasting.

3.2. Data processing

3.2.1. Decomposition process

The influence of noise on the data cannot be ignored when pre-processing the data, the data need to be noise-reduced at first. Crude oil price data contains some outliers that actually degrade the prediction performance. This paper decomposes the crude oil price signal into 7 IMFs. The VMD algorithm eliminates the high-frequency IMF (i.e., IMF7), which is considered a noise component. VMD's moderate bandwidth constraint and tolerance are set to 2000 and 10^{-7} , respectively. Fig. 6 shows the results of the VMD decomposition.

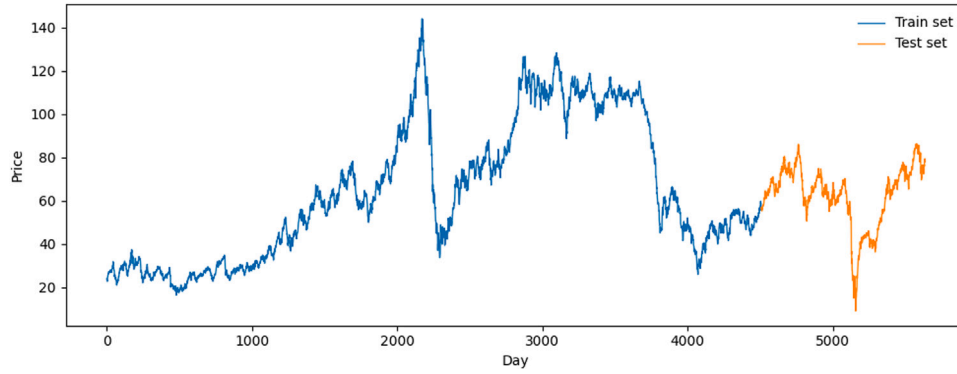


Fig. 5. Daily trend chart of Brent crude oil spot price.

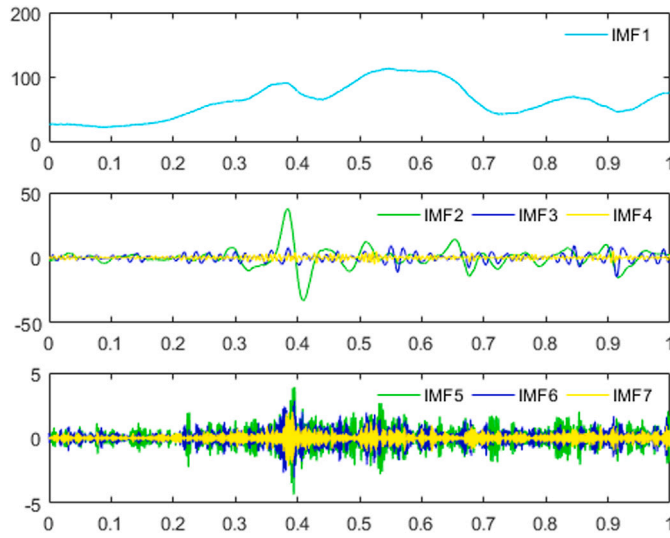


Fig. 6. The IMFs obtained using VMD.

3.2.2. Reconstruction process

One of the highlights of the model in this paper is that the reconstruction of the phase space is first performed during data processing. The outcomes of the autocorrelation coefficient method and the false nearest neighbors method when the phase space is reconstructed are displayed in Fig. 7. The autocorrelation function determines the value of τ to be 254. The rate of erroneous neighbors has reached its lowest point when the embedding dimension $m = 4$.

The RANSAC curve and average divergence curve are shown as the dashed line and the solid line, respectively, in Fig. 8. According to the small data sets method, the slope of the RANSAC curve is the maximum Lyapunov exponent. Small adjustments in crude oil prices will display exponential increases over time and results in massive changes, according to the maximal Lyapunov exponent $\lambda_1 = 0.0525$. This also demonstrates the crude oil price system's chaotic character.

At the same time, although the chaotic nature of the system makes its long-term prediction unreliable, the calculated maximum Lyapunov exponent is relatively small, which indicates that short-term prediction of the system evolution is still possible. According to the Lyapunov exponent properties, it is used to deal with the problem of quantitative predictability duration measurement. The maximum predictable time scale in chaotic time series is defined as the reciprocal of the system's

maximum Lyapunov exponent.

$$T = \frac{1}{\lambda_1}. \quad (16)$$

Given that it reflects the maximum amount of time required for the state error to double, it is one of the indicators of the accuracy of the prediction. The limit for computing the maximum prediction scale is $T = 19$. This indicates that the application of this chaotic system is valid for all data predictions within 19 days. In order to test the practicability of the model, this paper chooses to make multi-step forecasts of one day, one week, half a month and 19 days for the daily crude oil price series within the maximum forecast scale allowed by the chaotic conditions.

3.3. Forecasting components

3.3.1. Predictive model analysis

Phase space reconstruction restores a one-dimensional time series to a chaotic system, allowing prediction to be performed in a multi-dimensional phase space. The input time steps of the model are determined by m value which is 4. The hyperparameters of the prediction model are shown in Table 2. The combination of forecasting models determines forecasting outcomes. A CNN-BiLSTM hybrid model for forecasting crude oil prices for one day, one week, half a month, and 19 days is proposed in this study. To verify the effectiveness of the hybrid prediction model, we compare the prediction results with single LSTM, BiLSTM and CNN-LSTM hybrid model. Fig. 9 depicts the total predicted trend results. It can be seen that the prediction models show better results in terms of overall trends after phase space reconstruction of the data and after denoising. At the same time, it can be found that the prediction curves of the proposed model are closer to the true values at different prediction times, especially in one-day prediction when it almost coincides with the true curves. This shows that the proposed model can fully learn the important features in the data and exhibits good generalization ability.

The specific output of each model is shown in Figs. 10–13. One can generally identify the number of days the model has good and bad predictions by observing the prediction performance on the test set in the upper left graph. The upper right picture displays the results of the model's predictions over the last 60, 70, 80, and 90 days of price data from the test set, allowing for a more thorough examination of the model's prediction capabilities. The change in the loss function when the number of iterations is updated is plotted in the bottom left graph, evaluating the difference between the projected value and the actual value. The robustness of the model is improved via a reduced loss function. A residual plot of predicted against true values, shown in

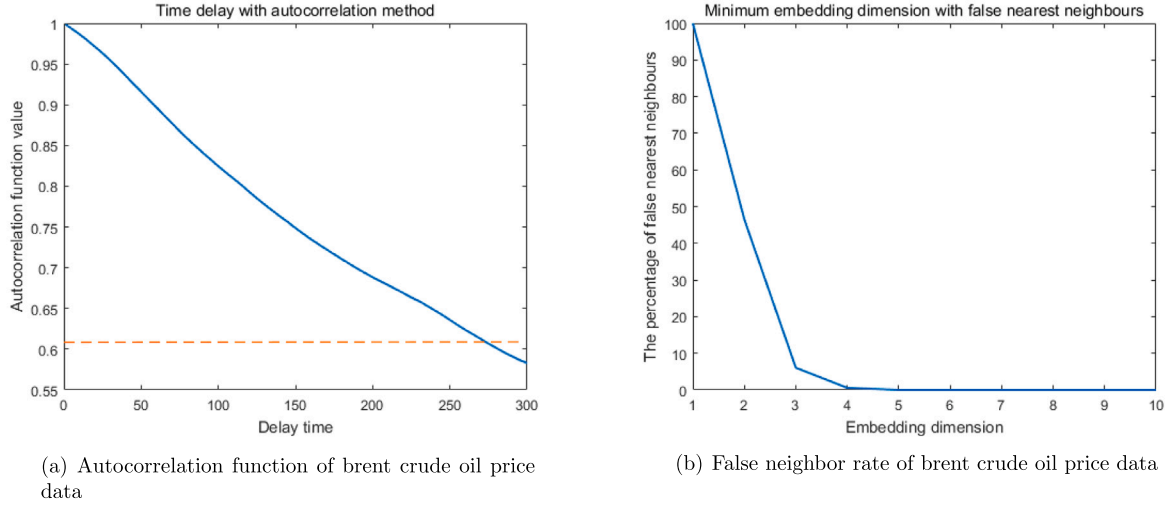


Fig. 7. The results of RSR.

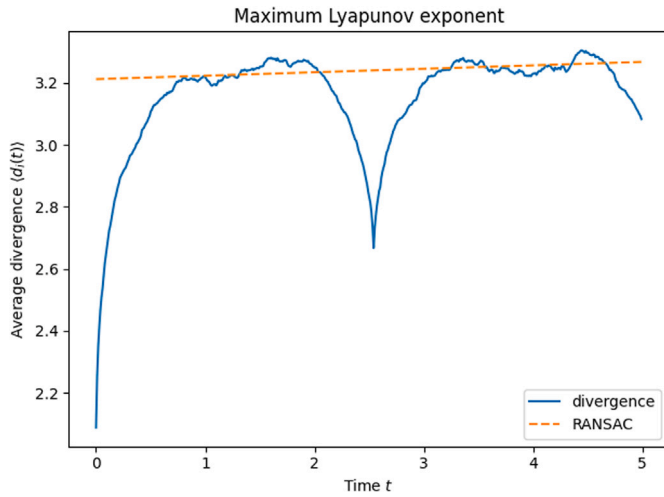


Fig. 8. The curve of average divergence.

Table 2
The hyperparameters of the prediction model.

Model parameters	Selected
Number of the filters in the CNN	64
Size of the kernel in the CNN	4
Number of neurons in LSTM	128
Number of neurons in BiLSTM	128
Iterations	20
Batch size	64
Activation function	ReLU
Optimizer	Adam

the bottom right graph, shows accuracy of the model anticipated values for each day. It can be seen from Figs. 10–13 that the prediction effect of the model declines as the time steps increase. But at each time step, the CNN-BiLSTM hybrid model shows its superiority.

3.3.2. Predictive result analysis

The mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) are

used to measure the prediction results of the models in this research in order to precisely evaluate their prediction accuracy. The better the model's training and the more appropriate the information mined, the smaller the order of magnitude of the four measures. The expressions are as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (17)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (18)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (19)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (20)$$

Among them, $\hat{y} = \{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n\}$ represents the predicted value, and $y = \{y_1, y_2, \dots, y_n\}$ denotes the real value. Furthermore, the directional statistic (Dstat) (Yao and Tan, 2000) which is an important indicator for price forecasting is also considered:

$$\text{Dstat} = \frac{1}{n} \sum_{i=1}^n \alpha_i \times 100\%, \quad (21)$$

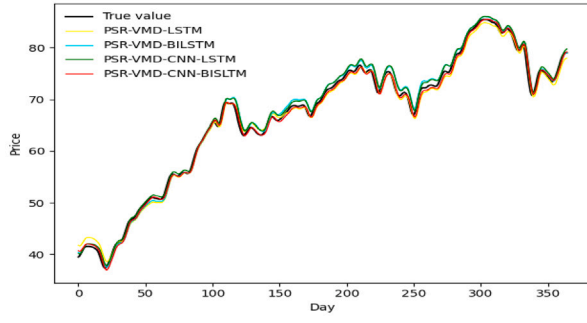
where

$$\alpha_i = \begin{cases} 1, & \text{if } (y_i - y_{i-1})(\hat{y}_i - \hat{y}_{i-1}) \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

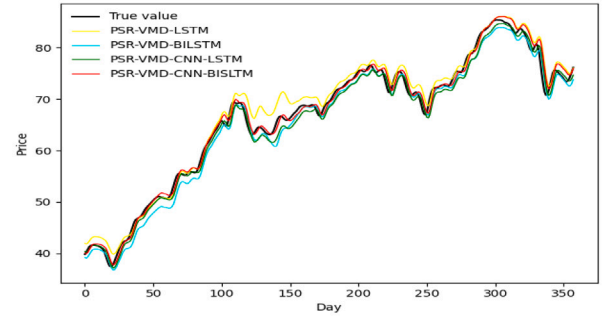
The better forecasting model will provide higher Dstat values.

This article focuses on forecasting the time series of crude oil prices. In terms of forecasting, this paper selects the deep learning models including LSTM, BiLSTM model CNN-LSTM model, and four state of the art (SOTA) deep learning models including PatchTST (Nie et al., 2022), Nbeats (Oreshkin et al., 2020), Temporal Fusion Transformer (TFT) (Lim et al., 2020) and Informer (Zhou et al., 2020) in time series forecasting as benchmarks for the CNN-BiLSTM model proposed in this paper.

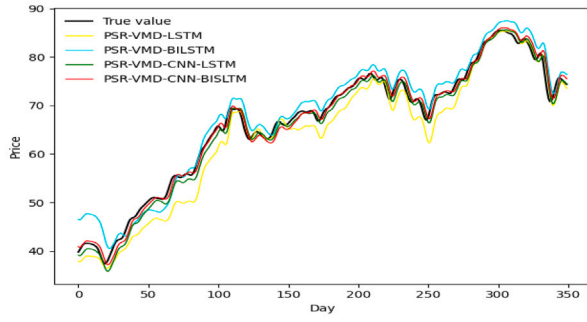
To verify the necessity of PSR and VMD and the superiority of the CNN-BiLSTM prediction model, the prediction results of the VMD-PSR-CNN-BiLSTM model will be compared to models with the same processing (VMD-PSR-LSTM, VMD-PSR-BiLSTM and VMD-PSR-CNN-LSTM



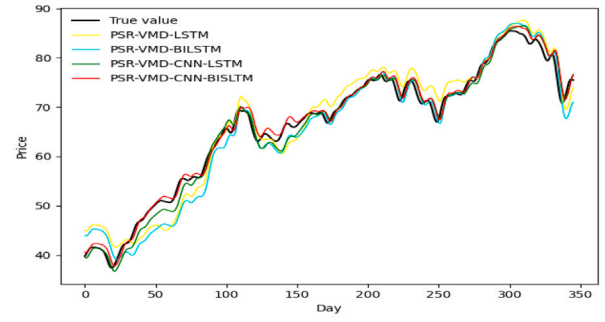
(a) Predicted trend of daily crude oil price (one day)



(b) Predicted trend of daily crude oil price (a week)

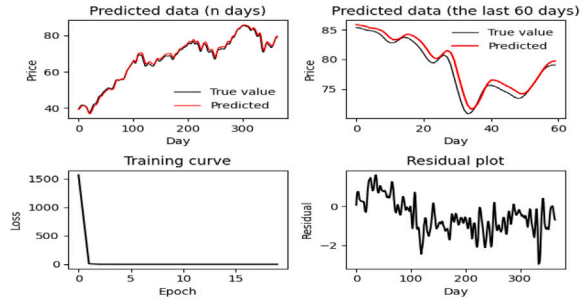


(c) Predicted trend of daily crude oil price (half a month)

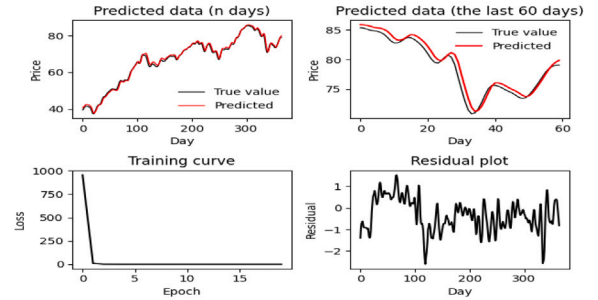


(d) Predicted trend of daily crude oil price (19 days)

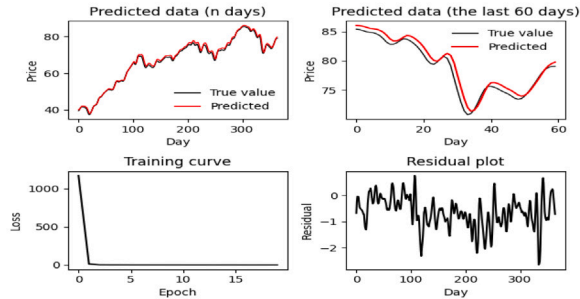
Fig. 9. The overall prediction trend of crude oil price.



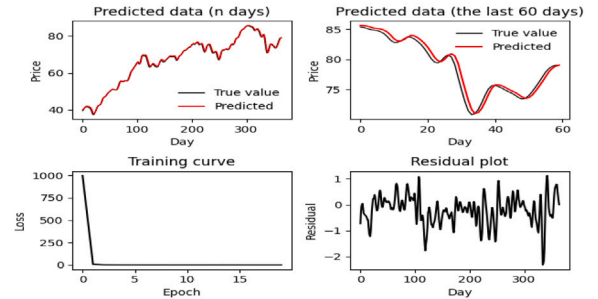
(a) The results of VMD-PSR-LSTM



(b) The results of VMD-PSR-BiLSTM

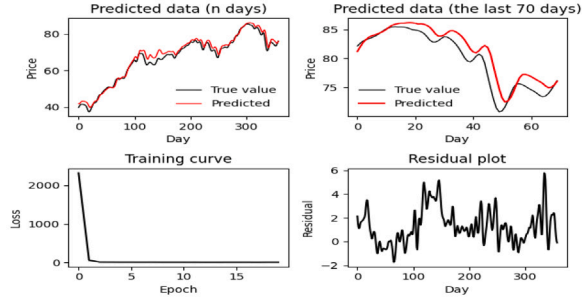


(c) The results of VMD-PSR-CNN-LSTM

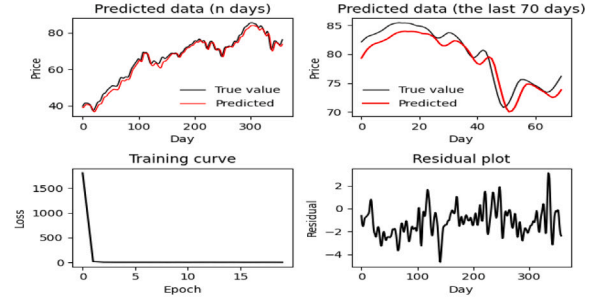


(d) The results of VMD-PSR-CNN-BiLSTM

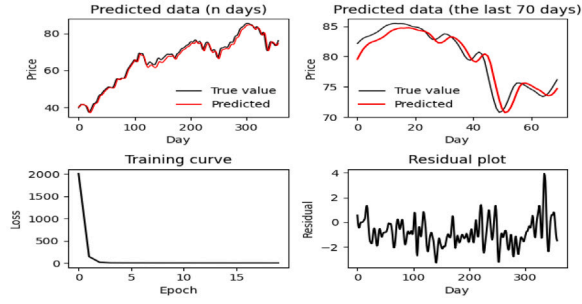
Fig. 10. The results of one day prediction with four models.



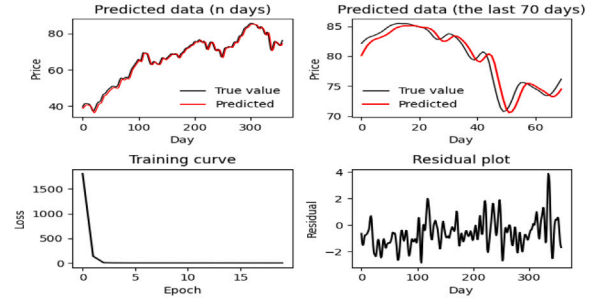
(a) The results of VMD-PSR-LSTM



(b) The results of VMD-PSR-BiLSTM

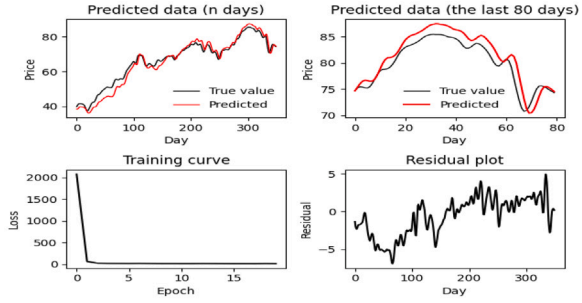


(c) The results of VMD-PSR-CNN-LSTM

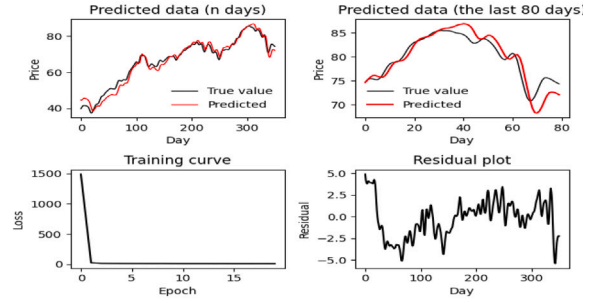


(d) The results of VMD-PSR-CNN-BiLSTM

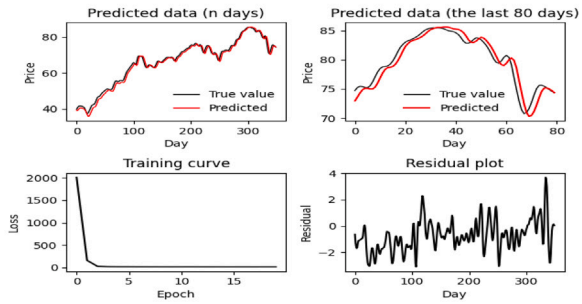
Fig. 11. The results of a week prediction with four models.



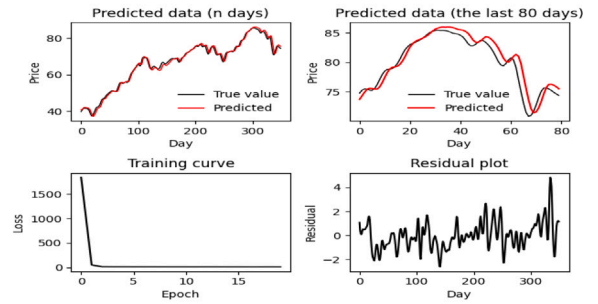
(a) The results of VMD-PSR-LSTM



(b) The results of VMD-PSR-BiLSTM



(c) The results of VMD-PSR-CNN-LSTM



(d) The results of VMD-PSR-CNN-BiLSTM

Fig. 12. The results of half a month prediction with four models.

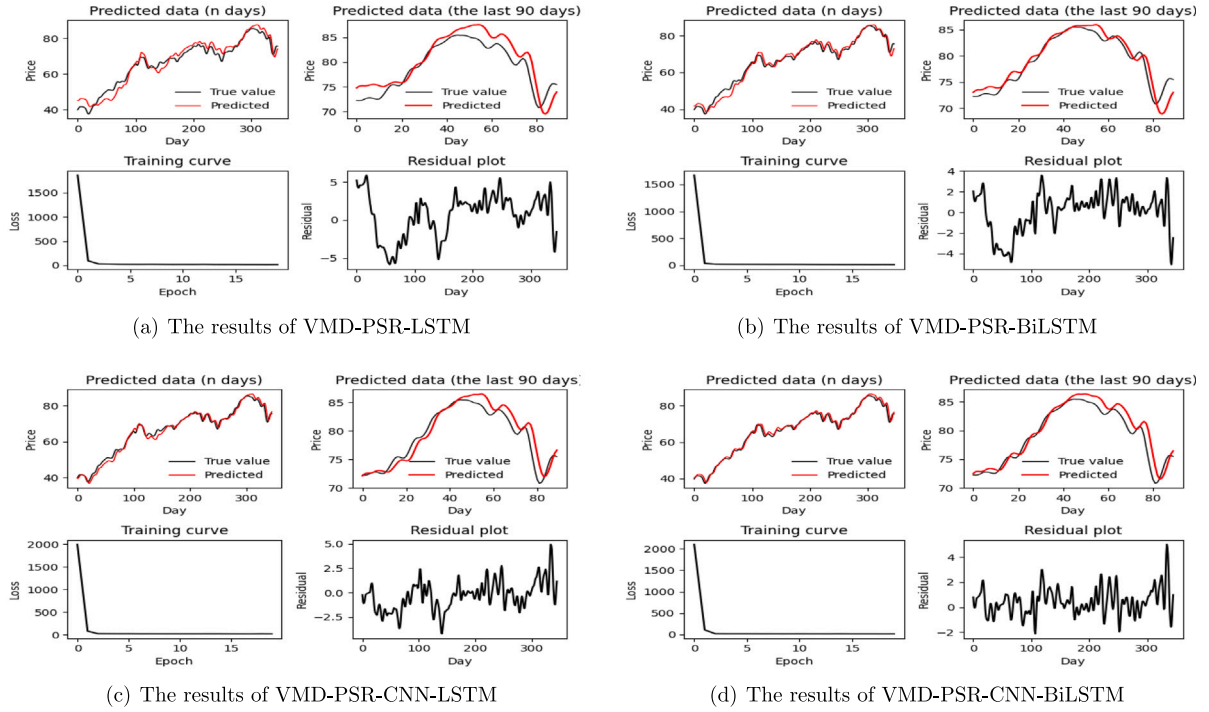


Fig. 13. The results of 19 days prediction with four models.

model), models with only reconstruction (PSR-LSTM, PSR-BiLSTM, PSR-CNN-LSTM and PSR-CNN-BiLSTM model), the model with only noise reduction (VMD-CNN-BiLSTM model) and the baseline models without any processing (LSTM, BiLSTM and CNN-BiLSTM model). In the unprocessed case, the inputs of forecasting models including SOTA models and LSTM, BiLSTM, an CNNBiLSTM are the original series. In the after configuration case, the inputs of LSTM, BiLSTM, CNNLSTM and CNNBiLSTM are the phase space reconstructed using the original series. In the noise reduction case, the inputs of CNN-BiLSTM are series processed by VMD. In the after processing case, the inputs of LSTM, BiLSTM, CNN-LSTM and CNN-BiLSTM are series phase space constructed using the series processed by VMD. Table 3 shows the prediction performance of each model in daily crude oil price data. The analysis of the results is provided as follows.

We first observe the results of the SOTA and unprocessed models, it can be seen that PatchTST model achieved the most accurate predictions in Cases 1,2 and 4. TFT has achieved the best results in Case 3. The combined model CNN-BiLSTM does not perform as well as LSTM and BiLSTM. There may be two reasons for this result. The first is that for complex one-dimensional time series data with huge fluctuations, data preprocessing is necessary. Secondly, it also shows that complex models do not always perform better than single models. The forecasting outcome depends on the model learning reasonably about the data features. For the SOTA models, the PatchTST, Nbeats, TFT and Informer models perform much better than LSTM, BiLSTM, and CNN-BiLSTM in Case 1,3,4. In Case 2, Nbeats and Informer do not perform as well as LSTM, BiLSTM and CNN-BiLSTM.

Observing the results of the models that have only been reconstructed (PSR-LSTM, PSR-BiLSTM, PSR-CNN-LSTM and PSR-CNN-BiLSTM model), it is obvious that this process not only improves the forecasting effect of each forecasting model at different time steps, but also the combined model starts to play an important role. This is because the reconstruction of the phase space makes the one-dimensional time series data expand into a multi-dimensional phase space, the learning space of the original data is expanded, and the phase space restores the spatial regularity and order of the chaotic original data. The CNN-BiLSTM model achieves the best learning of data

features in space and time, so it obtains the most accurate prediction results compared to other models. Noise removal is a common data processing operation. Observing the results of the VMD-CNN-BiLSTM model, the VMD algorithm also improves the prediction effect as a simple denoising tool instead of a decomposition tool in this paper.

In view of the above analysis, it is necessary to combine the prominent PSR method with the VMD algorithm to process the data. This paper builds VMD-PSR-LSTM, VMD-PSR-BiLSTM, VMD-PSR-CNN-LSTM and VMD-PSR-CNN-BiLSTM model, and it turns out that the accuracy of each prediction model has been improved. Among them, the prediction error of the VMD-PSR-CNN-BiLSTM model is the smallest and it provides the highest Dstat value. Furthermore, the predictive validity of the proposed model cannot be ignored. The predictions in this paper are limited to the short-term validity of the chaotic system, and multi-scale time-step predictions of one day, one week, half a month and 19 days are carried out on crude oil price data. The improvement effect of the PSR method is significant at each time step, while the VMD algorithm only has limited effect as a noise reduction tool. Overall, the predictive performance of each model decreases with increasing time steps. Especially under the maximum forecast of 19 days, the forecast results of the model are comparable. However, compared with other models, the proposed combined model can always achieve the best prediction results.

3.3.3. Diebold Mariano test

The Diebold Mariano (DM) test (Diebold and Mariano, 2002) is commonly used to compare the forecasting performance of two time series models and determine which model's forecasts are superior.

$$DM = \frac{\sum_{t=1}^T [L_t^A(y_t, \hat{y}_t) - L_t^B(y_t, \hat{y}_t)]}{\sqrt{S^2/p}} \quad (22)$$

where $L(\cdot)$ represents the loss function for forecasting errors, and S^2 is the sample variance defined by $d_t = L_t^A(y_t, \hat{y}_t) - L_t^B(y_t, \hat{y}_t)$. p denotes the degrees of freedom. Given the forecasts from model A and model B, conducting the test evaluates whether the loss difference of model A is significantly higher than or equal to that of model B. Evaluating the model performances using MSE, RMSE, MAE, MAPE, and other metrics

Table 3

Evaluation results of compared models in terms of MSE, RMSE, MAE, MAPE and Dstat.

	MODEL	MSE	RMSE	MAE	MAPE (%)	Dstat
Case 1 forecasting in one day						
SOTA	PatchTST	1.078	0.874	0.932	1.126	0.537
	Nbeats	2.024	1.022	0.805	1.969	0.541
	TFT	1.851	0.947	1.003	1.582	0.534
	Informer	1.489	1.339	0.998	1.363	0.532
Unprocessed	LSTM	2.132	1.460	1.068	2.044	0.607
	BiLSTM	2.153	1.467	1.025	1.975	0.602
	CNN-BiLSTM	2.425	1.557	1.098	2.092	0.604
	PSR-LSTM	2.286	1.512	1.121	1.790	0.648
After Reconfiguration	PSR-BiLSTM	1.849	1.360	0.970	1.506	0.643
	PSR-CNN-LSTM	1.794	1.339	0.943	1.488	0.637
	PSR-CNN-BiLSTM	1.714	1.309	0.928	1.451	0.640
After noise reduction	VMD-CNN-BiLSTM	0.516	0.718	0.504	1.028	0.731
	PSR-VMD-LSTM	0.962	0.981	0.814	1.260	0.843
After processing	VMD-PSR-BiLSTM	0.668	0.817	0.662	1.063	0.852
	VMD-PSR-CNN-LSTM	0.762	0.873	0.705	1.048	0.863
	VMD-PSR-CNN-BiLSTM	0.335	0.579	0.430	0.662	0.874
Case 2 forecasting in a week						
SOTA	PatchTST	6.392	2.789	1.898	3.391	0.504
	Nbeats	16.336	4.042	2.923	6.929	0.538
	TFT	6.637	2.576	1.852	3.804	0.539
	Informer	11.604	3.407	1.605	4.602	0.522
Unprocessed	LSTM	9.518	3.085	2.138	4.215	0.599
	BiLSTM	8.302	2.881	2.034	3.961	0.599
	CNN-BiLSTM	8.926	2.988	2.109	4.122	0.599
	PSR-LSTM	9.857	3.140	2.352	3.672	0.580
After Reconfiguration	PSR-BiLSTM	8.438	2.905	2.169	3.325	0.602
	PSR-CNN-LSTM	6.136	2.477	1.861	2.899	0.597
	PSR-CNN-BiLSTM	6.054	2.461	1.799	2.819	0.605
After noise reduction	VMD-CNN-BiLSTM	7.600	2.757	1.978	3.805	0.622
	VMD-PSR-LSTM	7.934	2.817	2.168	3.401	0.731
After processing	VMD-PSR-BiLSTM	5.725	2.393	1.970	3.109	0.739
	VMD-PSR-CNN-LSTM	4.689	2.166	1.708	2.613	0.745
	VMD-PSR-CNN-BiLSTM	3.942	1.986	1.465	2.283	0.756
Case 3 forecasting in half a month						
SOTA	PatchTST	14.627	4.052	2.610	4.841	0.523
	Nbeats	18.567	3.309	3.135	5.593	0.525
	TFT	13.436	4.131	3.028	4.971	0.509
	Informer	16.562	3.868	2.947	4.269	0.537
Unprocessed	LSTM	19.830	4.453	3.347	6.434	0.601
	BiLSTM	18.607	4.314	3.133	6.09	0.603
	CNN-BiLSTM	18.655	4.319	3.032	5.981	0.573
	PSR-LSTM	21.234	4.608	3.743	6.072	0.569
After Reconfiguration	PSR-BiLSTM	20.448	4.522	3.420	5.613	0.567
	PSR-CNN-LSTM	13.716	3.793	3.066	4.838	0.595
	PSR-CNN-BiLSTM	12.257	3.501	2.786	4.324	0.592
After noise reduction	VMD-CNN-BiLSTM	16.806	4.100	2.819	5.541	0.606
	VMD-PSR-LSTM	18.352	4.284	3.340	5.579	0.726
After processing	VMD-PSR-BiLSTM	16.429	4.053	3.029	5.125	0.692
	VMD-PSR-CNN-LSTM	11.197	3.346	2.668	4.249	0.724
	VMD-PSR-CNN-BiLSTM	10.047	3.17	2.441	3.829	0.740
Case 4 forecasting in 19 days						
SOTA	PatchTST	17.585	4.754	3.145	5.345	0.512
	Nbeats	23.521	4.849	3.310	6.102	0.497
	TFT	18.351	4.352	3.254	5.635	0.519
	Informer	19.004	4.000	3.286	5.786	0.526
Unprocessed	LSTM	25.608	5.060	3.467	6.990	0.577
	BiLSTM	23.611	4.859	3.459	6.835	0.612
	CNN-BiLSTM	24.603	4.960	3.410	6.846	0.594
	PSR-LSTM	25.927	5.092	4.105	6.561	0.629
After Reconfiguration	PSR-BiLSTM	28.763	5.363	4.103	6.717	0.577
	PSR-CNN-LSTM	18.276	4.275	3.496	5.415	0.603
	PSR-CNN-BiLSTM	17.519	4.186	3.305	5.152	0.614
After noise reduction	VMD-CNN-BiLSTM	23.207	4.817	3.157	6.350	0.626
	VMD-PSR-LSTM	21.994	4.690	3.735	6.053	0.693
After processing	VMD-PSR-BiLSTM	25.038	5.004	3.829	6.306	0.725
	VMD-PSR-CNN-LSTM	16.092	4.012	3.130	4.913	0.728
	VMD-PSR-CNN-BiLSTM	13.681	3.699	2.821	4.440	0.742

Table 4
The DM test results.

MODEL	One day	A week	Half a month	19 days
Case 1 DM test results with MSE				
PatchTST	8.478	5.658	8.444	9.972
Nbeats	9.079	8.462	10.775	13.823
TFT	8.051	6.396	14.112	10.836
Informers	8.363	6.311	12.444	8.866
LSTM	9.822	8.510	11.801	8.505
BiLSTM	7.762	6.494	10.648	7.111
CNN-BiLSTM	7.943	8.377	9.049	7.072
PSR-LSTM	8.454	12.367	12.225	13.656
PSR-BiLSTM	8.168	8.633	10.196	10.067
PSR-CNN-LSTM	7.305	5.882	8.299	8.662
PSR-CNN-BiLSTM	7.459	5.594	7.859	6.171
VMD-CNN-BiLSTM	9.528	9.306	6.533	7.777
VMD-PSR-LSTM	10.175	10.251	14.358	15.399
VMD-PSR-BiLSTM	8.559	7.635	10.617	10.901
VMD-PSR-CNN-LSTM	7.231	4.817	5.007	5.818
Case 2 DM test results with MAE				
PatchTST	11.091	11.854	14.097	7.279
Nbeats	13.044	9.054	13.157	9.330
TFT	10.303	2.925	16.291	10.288
Informers	11.598	2.142	14.097	12.358
LSTM	16.506	12.324	15.397	9.462
BiLSTM	12.662	8.857	13.528	7.629
CNN-BiLSTM	12.488	10.541	11.003	8.857
PSR-LSTM	12.184	16.826	14.523	21.618
PSR-BiLSTM	10.406	10.925	12.505	12.183
PSR-CNN-LSTM	8.587	8.083	11.229	10.353
PSR-CNN-BiLSTM	7.696	7.411	10.074	8.060
VMD-CNN-BiLSTM	12.736	10.832	7.808	12.160
VMD-PSR-LSTM	11.090	12.391	16.43	20.039
VMD-PSR-BiLSTM	9.268	10.496	15.527	12.595
VMD-PSR-CNN-LSTM	6.887	7.322	6.742	5.702
Case 3 DM test results with MAPE				
PatchTST	10.022	8.335	12.083	9.139
Nbeats	10.020	10.064	11.067	10.072
TFT	9.026	10.709	13.111	9.098
Informers	12.030	12.057	12.436	12.048
LSTM	16.240	12.483	16.202	9.438
BiLSTM	12.638	9.014	14.169	7.677
CNN-BiLSTM	12.358	10.980	11.213	8.944
PSR-LSTM	12.344	17.283	13.759	21.151
PSR-BiLSTM	10.628	11.296	11.973	11.386
PSR-CNN-LSTM	8.166	8.429	11.368	10.273
PSR-CNN-BiLSTM	7.457	7.860	9.722	8.001
VMD-CNN-BiLSTM	12.086	11.247	6.869	12.052
VMD-PSR-LSTM	9.275	12.158	16.487	17.021
VMD-PSR-BiLSTM	9.075	10.906	11.959	12.423
VMD-PSR-CNN-LSTM	5.608	7.448	7.403	5.817

can be a good way to assess the model's prediction accuracy. The loss function will be set as MSE, MAE, and MAPE for DM detection. Table 4 displays the results of the DM test.

As shown in Table 4, the DM values of the proposed VMD-PSR-CNN-BiLSTM model is larger than the cutoff value $z_{0.01/2} = 2.58$ which denotes the z-value in the normal distribution table at the significance level of 1% with all the comparative models when forecasting crude oil price data for one day, one week, half a month and 19 days with the loss functions given by MAE, MSE, and MAPE, respectively. For example, the DM test values of the proposed model and other comparison models are much greater than 2.58. In general, there are substantial differences between the suggested forecasting model and other comparison models at 1% significance level. Consequently, the proposed model achieves a highly competitive prediction accuracy. To show the advantages of VMD and PSR, Table 5 shows the DM test between each experiment with and without VMD and PSR. It is observed that the applications

of PSR and VMD are effective in boosting forecasting accuracy. For instance, in Case 4, the DM test results indicate that the forecasting efficiency between BiLSTM, PSR-BiLSTM and VMD-PSR-BiLSTM are different. In Case 2, the DM test results also indicate that LSTM, PSR-LSTM and VMD-PSR-LSTM have shown different forecasting efficiency.

4. Conclusion

Crude oil prices are easily affected by objective and sudden human factors, with huge fluctuations and extremely difficult forecasting. It has been a topic of exploration significance since ancient times. Especially since COVID-19 pandemic occurs, accurately grasping its development trend has important strategic significance for the economic development of various countries. The research shows that the volatility and complexity of crude oil price series make crude oil price a nonlinear dynamic system with chaotic properties. Chaos theory should

Table 5

DM tests between each experiment with and without PSR/VMD using MSE as loss function.

Models	Case 1		Case 2	
	PSR-LSTM	VMD-PSR-LSTM	PSR-LSTM	VMD-PSR-LSTM
LSTM	1.4481	7.1847***	−8.4704***	−3.7992***
PSR-LSTM	/	5.5647***	/	5.7373***
Models	PSR-BiLSTM		PSR-BiLSTM	
	PSR-BiLSTM	VMD-PSR-BiLSTM	PSR-BiLSTM	VMD-PSR-BiLSTM
BiLSTM	3.4230***	5.8447***	−5.6497***	−3.1514**
PSR-BiLSTM	/	4.0510***	/	0.1629
Models	PSR-CNN-BiLSTM		PSR-CNN-BiLSTM	
	PSR-CNN-BiLSTM	VMD-PSR-CNNBiLSTM	PSR-CNN-BiLSTM	VMD-PSR-CNNBiLSTM
CNN-BiLSTM	5.2326***	7.8692***	1.6117	8.3769***
PSR-CNN-BiLSTM	/	7.2957***	/	5.5935***
Models	Case 3		Case 4	
	PSR-LSTM	VMD-PSR-LSTM	PSR-LSTM	VMD-PSR-LSTM
LSTM	−8.8908***	−10.1051***	−9.0595***	−10.4475***
PSR-LSTM	/	1.7458*	/	−1.4636
Models	PSR-BiLSTM		PSR-BiLSTM	
	PSR-BiLSTM	VMD-PSR-BiLSTM	PSR-BiLSTM	VMD-PSR-BiLSTM
BiLSTM	−6.1116***	−5.0237***	−7.7167***	−9.1162***
PSR-BiLSTM	/	0.1239	/	−4.1791***
Models	PSR-CNN-BiLSTM		PSR-CNN-BiLSTM	
	PSR-CNN-BiLSTM	VMD-PSR-CNNBiLSTM	PSR-CNN-BiLSTM	VMD-PSR-CNNBiLSTM
CNN-BiLSTM	−1.0740	9.0494***	−0.1928	7.0719***
PSR-CNN-BiLSTM	/	7.8594***	/	6.1705***

*** Represents 0.01 significance level.

** Represents 0.5 significance level.

* Represents 0.1 significance level.

be used to study crude oil prices from complexity science. Based on this view, this paper uses empirical data of daily spot prices of Brent crude oil from January 4, 2000, to December 31, 2021, to make price forecasts.

Through the analysis and summary of the empirical results of the proposed VMD-PSR-CNN-BiLSTM model, the conclusions of this paper are listed as follows: first, the VMD method is selected as the noise reduction tool. It reduces the nonlinear and non-stationary characteristics of crude oil price data and the empirical results show that the prediction accuracy is greatly improved after removing high-frequency noise. Secondly, we find evidence and necessity for the effective application of chaos theory in crude oil price series. The introduction of the phase space reconstruction technique enables the prediction to be performed in the phase space. This paper uses the maximum Lyapunov exponent to verify the chaos of crude oil price series and reconstruct the phase space. Meanwhile, the multi-step prediction at one day, a week, half a month, and 19 days prediction time steps of crude oil price data is carried out. In addition to verifying the effectiveness of the model on different prediction scales, the effectiveness of chaos theory in its applicable range is also explored. Thirdly, a hybrid model of the appropriate deep learning method is used for the prediction part. In phase space, data has both temporal and spatial characteristics. CNN is used to obtain the spatial characteristics of the phase space, and BiLSTM is used to obtain the spatio-temporal characteristics under the spatial characteristics. In comparison to other unreconstructed models, unreduced noise models, and benchmark models, the prediction results of the proposed model demonstrate higher prediction accuracy. Besides, the results of the DM test further demonstrate the model's significant advantage over competing models. It fully demonstrates the effectiveness and extensive research potential of the proposed model in forecasting crude oil price series.

Considering the excellent performance of the suggested method, it can assist essential personnel and financial investors in taking timely and acceptable actions to deal with continually fluctuating crude oil prices, thereby lessening the economic impact of global oil price variations on countries. Other chaotic time series can also be analyzed using the approach. Most of the parameters in this paper are based on manual settings, and subsequent research work will further investigate the optimization of hyperparameters in the prediction model. Besides,

this paper provides effective point forecasts for crude oil price data, interval forecasts and probabilistic forecasts will be the future research directions.

CRediT authorship contribution statement

Yao Dong: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **He Jiang:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Funding acquisition, Formal analysis, Data curation. **Yunting Guo:** Supervision, Software, Resources, Formal analysis. **Jianzhou Wang:** Visualization, Validation, Supervision, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

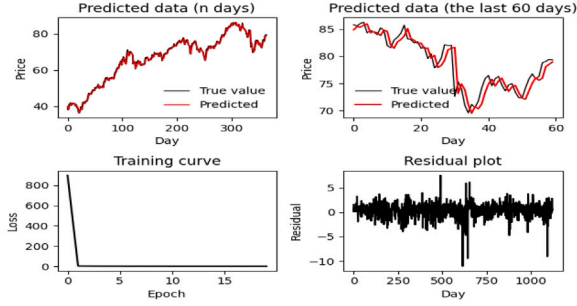
Data will be made available on request.

Acknowledgments

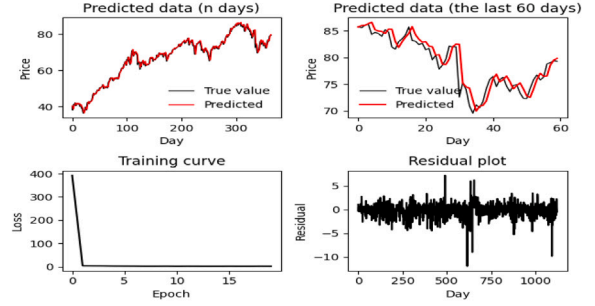
This research is supported by the National Natural Science Foundation of China (grant Nos. 72261014, 72371196, 71901109, 71861012), Natural Science Foundation of Jiangxi, China (20224BAB201007), Humanities and Social Sciences Foundation of Jiangxi Province, China (Grant No. TJ22201).

Appendix. Results of other models

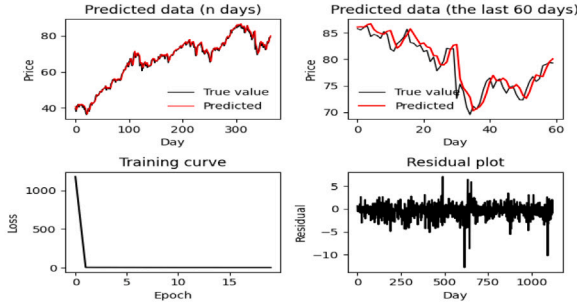
The specific predict results of the PSR-LSTM, PSR-BiLSTM, PSR-CNN-LSTM, PSR-CNN-BiLSTM models without noise reduction, the unreconstructed VMD-CNN-BiLSTM model, and the benchmarks models without any processing of the data, including LSTM, BiLSTM and CNN BiLSTM models in one day, a week, half a month and 19 days (see Figs. A.14–A.17).



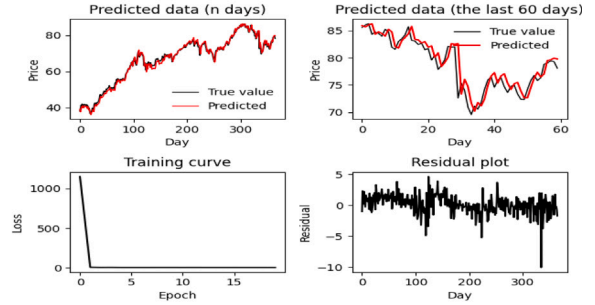
(a) The results of LSTM



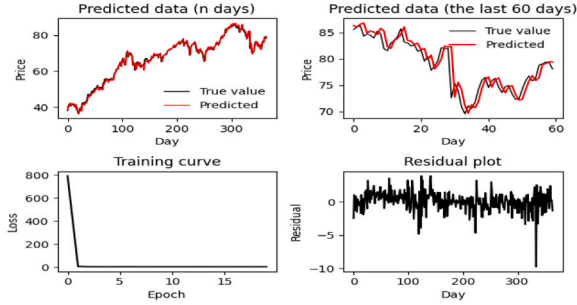
(b) The results of BiLSTM



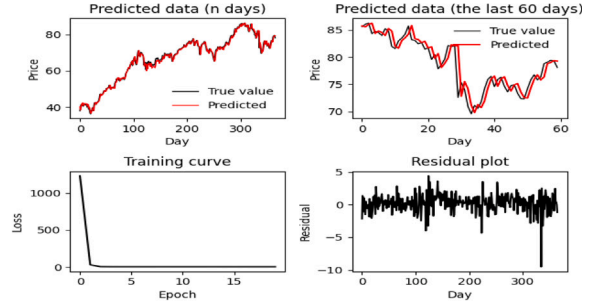
(c) The results of CNN-BiLSTM



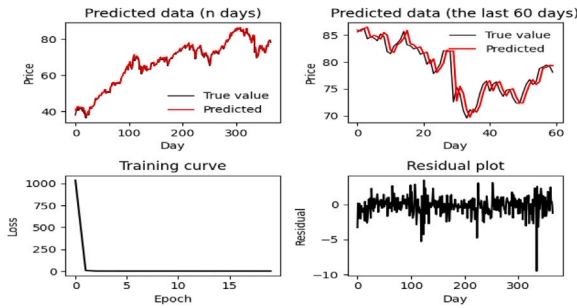
(d) The results of PSR-LSTM



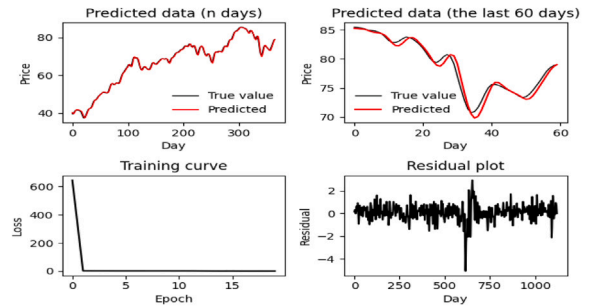
(e) The results of PSR-BiLSTM



(f) The results of PSR-CNN-LSTM

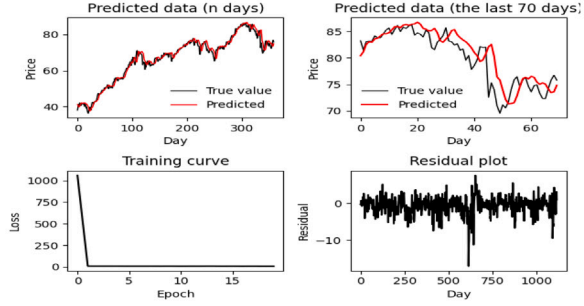


(g) The results of PSR-CNN-BiLSTM

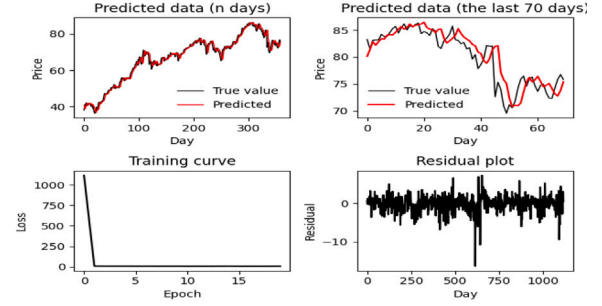


(h) The results of VMD-CNN-BiLSTM

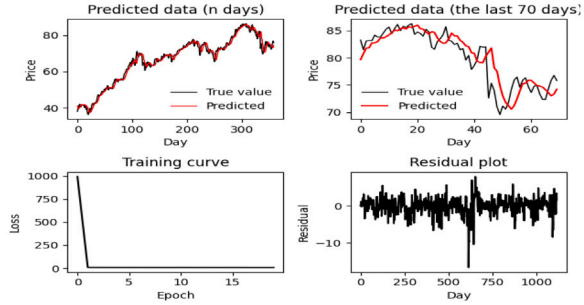
Fig. A.14. The results of one day prediction with eight models.



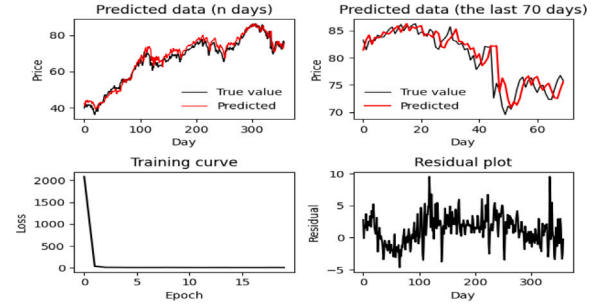
(a) The results of LSTM



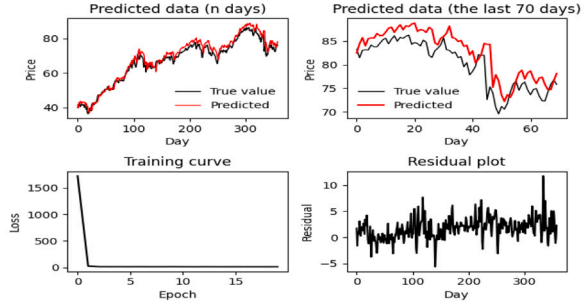
(b) The results of BiLSTM



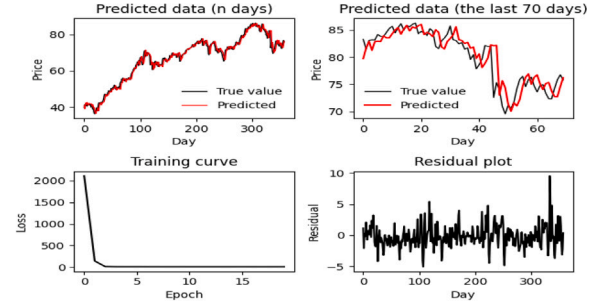
(c) The results of CNN-BiLSTM



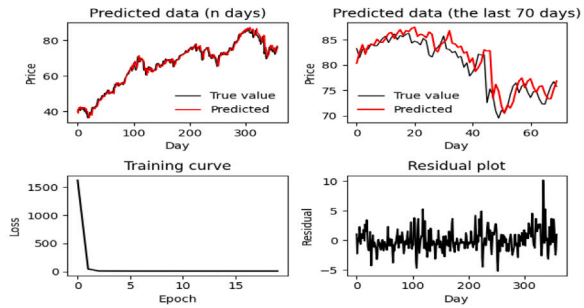
(d) The results of PSR-LSTM



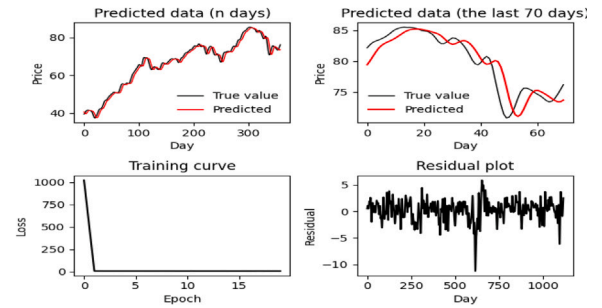
(e) The results of PSR-BiLSTM



(f) The results of PSR-CNN-LSTM

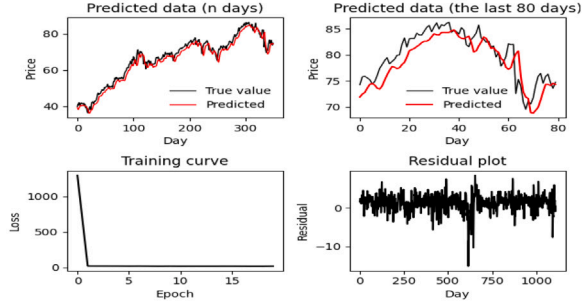


(g) The results of PSR-CNN-BiLSTM

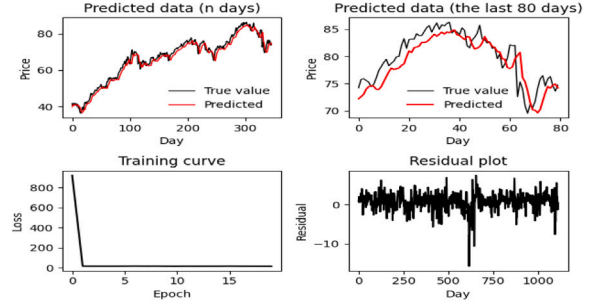


(h) The results of VMD-CNN-BiLSTM

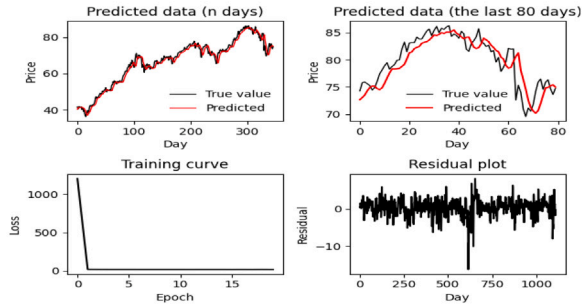
Fig. A.15. The results of a week prediction with eight models.



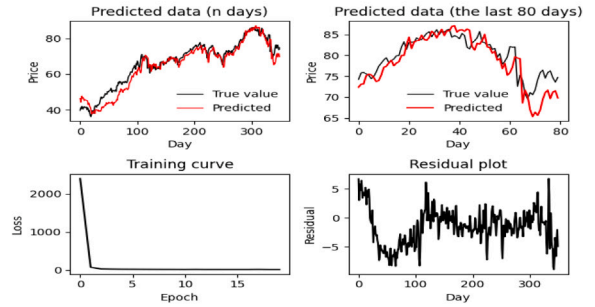
(a) The results of LSTM



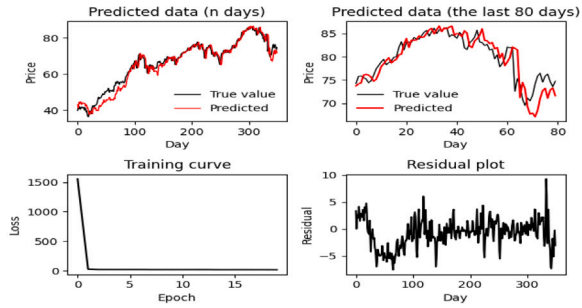
(b) The results of BiLSTM



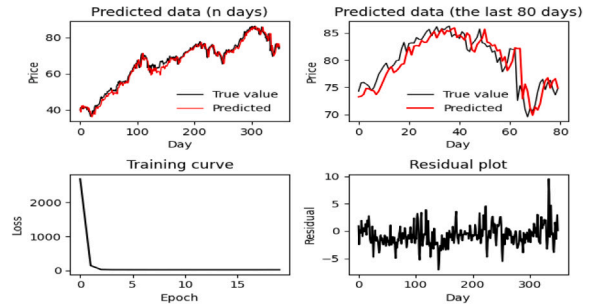
(c) The results of CNN-BiLSTM



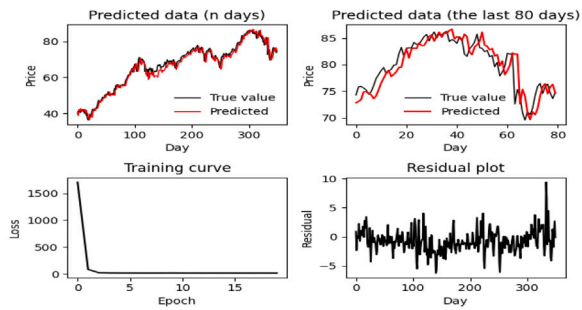
(d) The results of PSR-LSTM



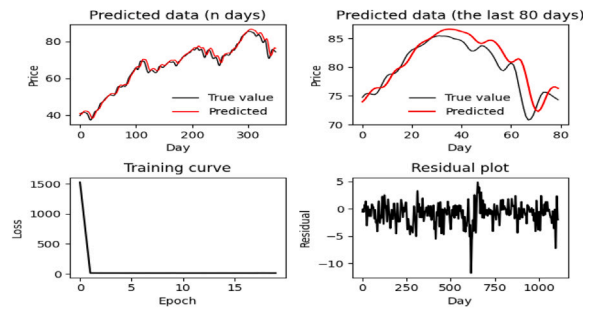
(e) The results of PSR-BiLSTM



(f) The results of PSR-CNN-LSTM



(g) The results of PSR-CNN-BiLSTM



(h) The results of VMD-CNN-BiLSTM

Fig. A.16. The results of half a month prediction with eight models.

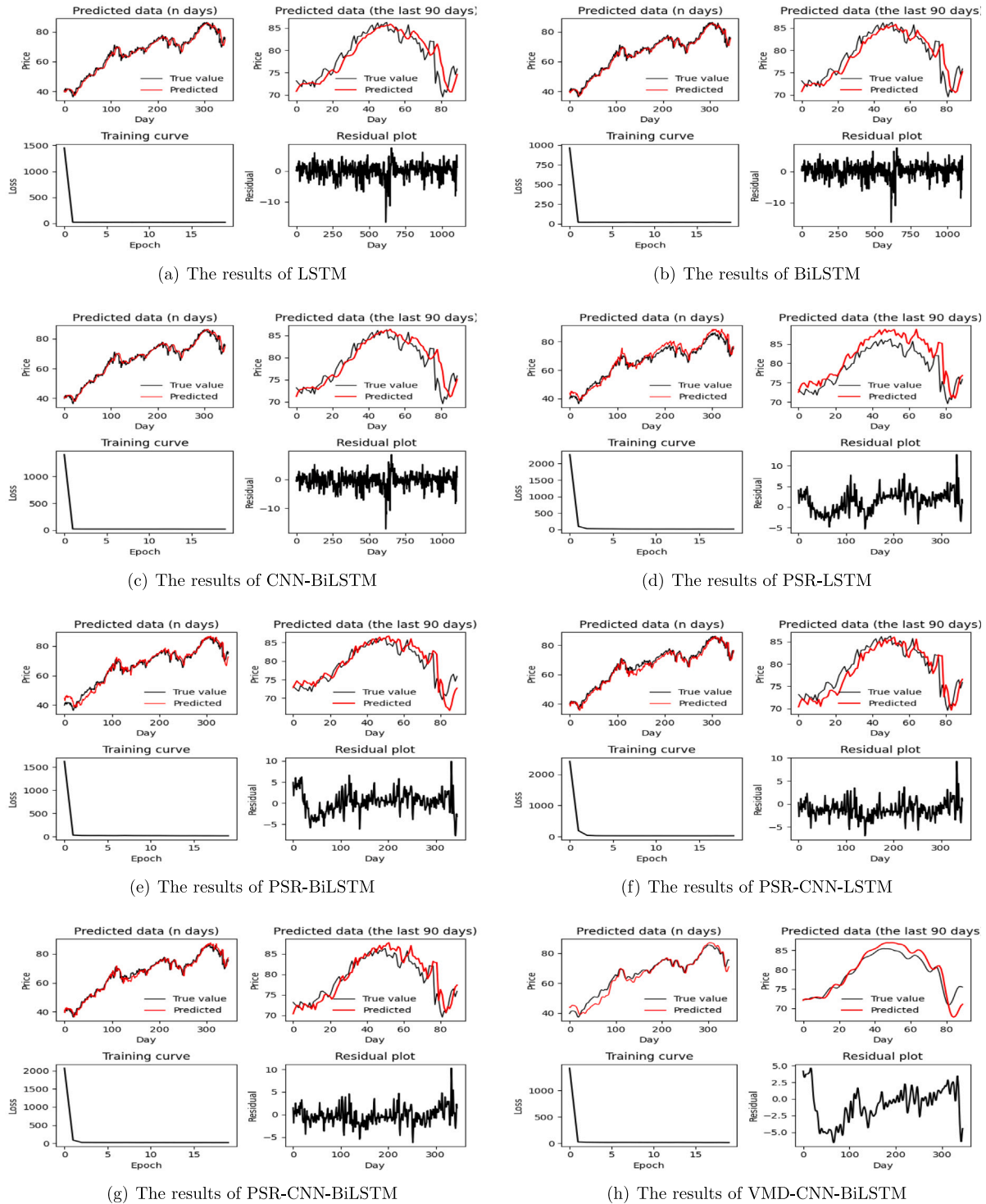


Fig. A.17. The results of 19 days prediction with eight models.

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